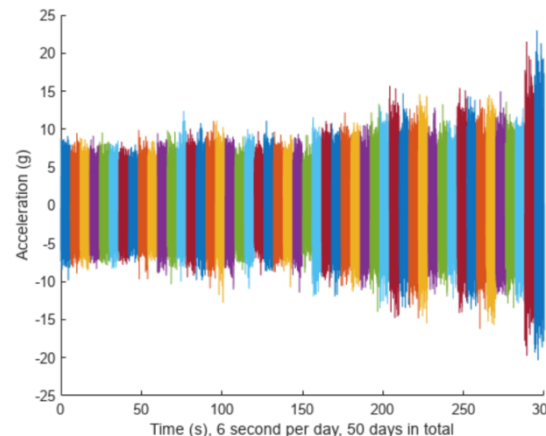


Project Description



- A historical run-to-failure dataset of a wind turbine bearing will be provided.
- The dataset consists of vibration signal and tachometer signal, which were acquired each day for 50 consecutive days (7 March – 25 April).
- An inner race fault developed and caused the failure of the bearing across the 50-day period.
- The sampling frequency of the vibration data is 97656 Hz. PPR=2 for tach signal.
- Dataset:
https://drive.google.com/drive/folders/1I1MXpZPy2LLYTxbYa6sJ3eXO4CRPVvJ1?usp=drive_link



Develop a Real-time Monitoring System for an Industrial Bearing using a Digital Twin Framework

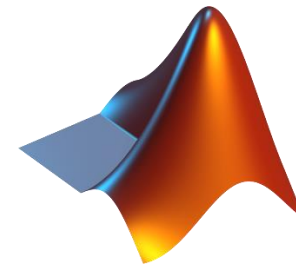
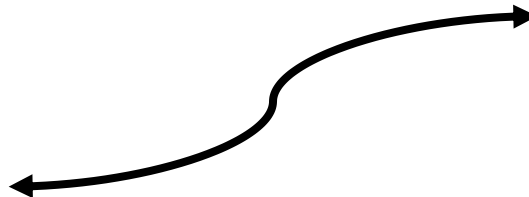
Project Description



- The Raspberry Pi will store the data.
- Use the data to simulate a real-time data collection and monitoring mechanism.
- Process the data to extract bearing health information.
- Indicate the real-time health of the bearing using a virtual model (a representative 3D bearing model that at least changes colour with change in its health condition) that is updated in real-time. The virtual model should also have other service capabilities (like remaining useful life, etc.)



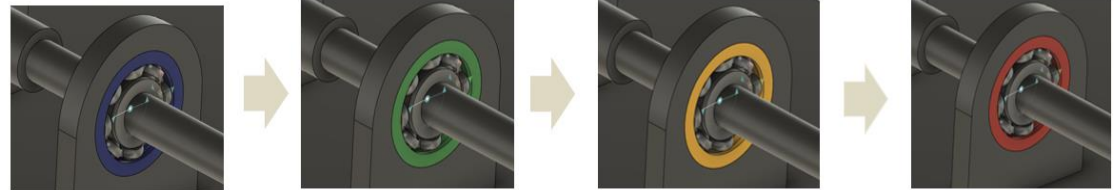
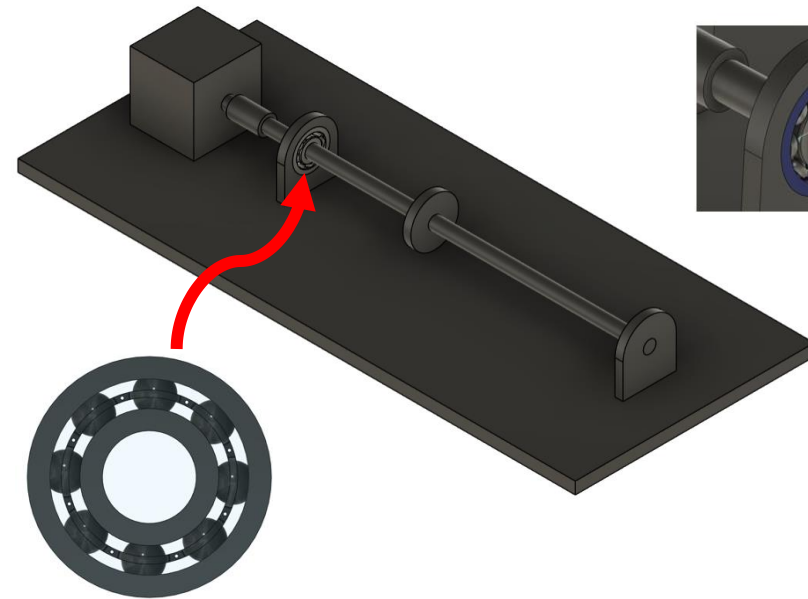
Data collection and processing



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Visualization and Other Services

Project Description



- Discuss feature engineering, how HI is created, compare different HI models, explain how the best model was identified etc.
- Clearly discuss how communication is set-up.
- Discuss how update is being performed, RUL etc.
- Explain main codes.
- Main report body should be of 10 pages (excluding Title, Table of Content, List of References etc.)
- Any additional information that helps you to explain your work in the main report, should be put as Appendix.

Tentative Marking Scheme:

- Report - 10 marks
- Code, Model etc. – 5 marks
- **Presentation & Real-time monitoring Demonstration – 5 marks (27 April 2026)**